MulBos: A Monte Carlo program for polarized multiboson production processes – Manual for Version 1.0

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Abstract

MulBos (for MultiBoson production) is a parton-level program to calculate polarized cross sections for W^+W^-, $W^\pm Z$, and ZZ production processes with fully leptonic decays at the LHC. The calculations are performed at next-to-leading order (NLO) QCD+EW accuracy. In this manual, we explain how to install and run the program. Important practical details are also provided. Extension to triboson production processes will be added in the future.

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1 Installation and run

The source code of MulBos can be downloaded at

`https://github.com/mulbos/mulbos`

1.1 Prerequisites

The basic installation requires GNU `make`, the Fortran compiler Intel `ifx`. A `C` compiler is required for LoopInts, which has been tested with `gcc`. The library LHAPDF[1] [1] is needed for parton distribution functions (PDF).

For easy compilation and run of the program, the simple python scripts `setup.py` and `run_VV.py` (with `VV=WZ, WpWm, ZZ`, located in `run_templates`) are provided. To run these scripts, `python3` is needed. Please note that LHAPDF also requires `python3` for installation.

1.2 Compilation and run

Upon downloading MulBos, one goes to the home directory of the program. From here, the compilation can be done with one command:

```
python3 setup.py
```

This command installs all necessary libraries, except for the LHAPDF. LHAPDF (version $\geq$ 6.2.3 is recommended) must be installed separately and the path to the installation folder must be provided in the `run_VV.py`.

The current version of MulBos is capable of calculating all polarized cross sections for the 4 diboson processes ZZ, W^+W^-, W^+Z, and W^-Z at NLO QCD+EW at proton-proton colliders. Any kinematic distribution can be generated. Event generation is not yet available.

Instructions to calculate polarized cross sections at NLO are given in Section 8. Here we just explain how to perform a run.

To perform a calculation for a specific process, say ZZ, one first creates a personal working directory named e.g. `my_working`. This directory can be placed anywhere. A good place is the MulBos home directory. One then copy the folder `run_templates/ZZ` to this working directory. Go to the folder `my_working/ZZ` and edit the file `run_ZZ.py` to provide the path to MulBos home directory (search `home_mulbos`) and the path to LHAPDF library (search `LHAPDF`). Then execute the following commands:

```
python3 run_ZZ.py
```

You can change the run parameters in `def run()`, the input parameters in `cards/input.dat.in`, and the cut parameters in `cards/cuts.dat.in`. These input and cut parameters have been chosen to be the same as Ref. [2] for the sake of comparison.

Upon running the `run_ZZ.py` script, a new folder named `results_*` (which is the default name, you can change it via the variable `output_dir` in `run_ZZ.py`) is created if it does not yet exist. The results are then stored in this directory. Folders named `run_IDp_IDc_polXY_*` are created [2], where `IDp` and `IDc` are the process and contribution

[1] `https://www.lhapdf.org/`

[2] One can easily change this name in `run_ZZ.py`.

identification numbers (ID), respectively. The list of these identification numbers are provided in Tables 2, 3, 4, 5, 6, 7, 8, 9, 10, 11. The last index `polXY` gives the polarization state, see Table 1 for the list of polarization IDs (`XY=IDh`).

In the `run_IDp_IDc_polXY_*` folder, the following files are created:

1. `makefile`
2. `main.F`
3. `input.dat`
4. `cuts.dat`
5. `mulbos`
6. `xs.txt`
7. `dist/`

This `run_IDp_IDc_polXY_*` directory is a complete program in itself. One can copy this folder to another place and compile it to create the executable `mulbos`, and then submit job from there. For an easy use, this compilation and job submission steps have been done in `run_ZZ.py`. After the job has finished, the integrated cross section is provided in the file `xs.txt`, where some basic information about the input parameters and integration parameters are also given. Kinematic distributions are provided in the `dist` folder. The current version of the program cannot generate events.

To get the full NLO results, one has to sum over all relevant processes (e.g. $\bar{U}U$, $\bar{D}D$, gg, $\gamma\gamma$, $b\bar{b}$) and contributions (LO, virtual corrections, real emission corrections, ...) for each distribution (or integrated cross section). See Section 8 for specific instructions.

1.3 Bug reports

If you encounter any problems, please inform us via `leducninh@gmail.com` or open an issue in github: `https://github.com/mulbos/mulbos/issues`.

2 Source and directory layout

The MulBos source tree contains the following subdirectories:

- `driver/`: Containing the main program `main.F` and a makefile template `makefile.in` which is used to create the final makefile (located in `run_IDp_IDc_polXY` as above discussed).
- `tools/`:
  - `utilities/`: Containing routines for calculating model parameters, run parameters, generating momenta of external-state particles, on-shell mappings, kinematic cuts, PDFs, filling histograms of kinematic distributions, the integrands of various contributions (LO, virtual corrections, real-emission corrections, ...). The Catani-Seymour-Dittmaier dipole subtraction method is here implemented. Renormalization constants for virtual corrections are here included. Routines to read in input files and print out important information are placed here as well.

  - `FormCalc_Tools/`: These interface files are needed for the calculation of the FormCalc matrix elements in the `processes` directory. They have been adapted from FormCalc-6.0. In more details, the purpose of these routines are to calculate spinors, polarization vectors for each external-state particle, and some basic functions of these quantities.
  - `integrators/`: Containing the Monte-Carlo (MC) integrator `BASES-5.1` [3, 4], which is needed for the phase-space integration and kinematic distribution generation.
  - `loops/`: Containing the library LoopInts, for the calculation of one-loop integrals needed for the virtual corrections or loop-induced processes.

- `processes/`: Routines to calculate helicity amplitudes for the processes provided. The production and decay helicity amplitudes are generated by FormCalc + FeynArts. Additional work is needed to combine these amplitudes according to the PA to obtained multi-gauge boson polarized amplitudes. FormCalc-6.0 and FeynArts-3.4 have been used to generate all amplitudes for the diboson processes. Importantly, the list of processes, contributions and their IDs are provided in the `processes_list.py` file located in this folder.
- `run_templates/`: Run templates for all processes.
- `manual/`: The Latex source files of this manual.

3 License

MulBos is distributed under the GNU General Public License (GPL) version 3. This ensures that the source code will be available to users, grants them the freedom to use and modify the program and sets out the conditions under which it can be redistributed. More details are provided in the `COPYING` file shipped together with the source code.

4 Credits

MulBos contains codes taken and adapted from other open-source programs. The matrix elements are based on automatically generated code from FeynArts+FormCalc [5–8]. Some main interface routines for process and input-parameter initialization are adapted from FormCalc-6.0 codes. The phase space generation codes, which generate off-shell momenta from random numbers, are based on codes taken from VBFNLO [9–11]. The routines which read in input parameter values from the card files are also taken from VBFNLO.

A very slightly modified version of the BASES-5.1 [3, 4] library is shipped together with the code. This is needed for Monte-Carlo integration and the generation of kinematic distributions.

One-loop integrals are calculated by the library LoopInts. Here, some codes are taken and adapted from LoopTools [6]. Specifically, they include the cache system (see `cache.c`), Passarino-Veltman reduction for 3-point and 4-point tensor integrals. The routines to solve the system of linear equations needed by this reduction method are also taken from LoopTools, see `solve-LU.F`. The core parts of LoopInts, including 2-point

integrals, 3-point and 4-point scalar integrals have been implemented by us from scratch. These codes have been cross checked and used in many calculations [12–17].

5 Code flow

After executing `python run_VV.py`, a new run folder named `run_*` is created in the `results_*` directory.

Go to this run folder, one finds the main program `main.F`, which reads in the input files (`input.dat` and `cuts.dat` located in the same directory), initializes important run parameters (e.g. colliding energy, model parameters) and performs the phase space integration. There is a `makefile` there, which links the the main program with the needed libraries.

The phase space integration, which is done via the subroutine `IntegratedCS` located in `tools/utilities/xsection.F`. Here, the number of integration variables is automatically calculated and the subroutine `BASES2` is called. This is an interface to the program `BASES` [4], which is provided in `tools/integrators/basesv5.1`. `BASES` reads in the integrand (named `FUNC`) and then performs the MC integration. It returns the integrated cross section (the mean value and the standard deviation) and a set of kinematic distributions in the folder `dist`.

The value of the integrand `FUNC` is calculated from the list of random variables $x(i)$, which are generated by `BASES`. The main subroutine here is `DifferentialCS` (located in `tools/utilities/Integrand.F`) where all the external momenta, PDFs, helicity amplitudes are calculated. Kinematic cuts and histogram filling are done via this subroutine as well.

6 Input files and parameters

MulBos reads in input parameters through the following input files:

- `run_VV.py`: Some run parameters.
- `input.dat.in`: Run and model parameters.
- `cuts.dat.in`: Kinematic cut parameters.

7 Checks

Extensive checks for all components have been done by consistency checks (e.g. UV and IR finiteness) as well as comparing with external programs (VBFNLO, Sherpa [18], MoCaNLO [19]).

8 Calculating polarized cross sections and distributions

Here we give instructions to calculate polarized cross sections and distributions at NLO QCD+EW. The list of Polarization ID numbers, denoted as IDh (h for helicity), are

provided in Table 1. The list of Process ID (IDp) and Contribution ID (IDc) numbers are provided in Table 2 for W^+Z; Table 3 for W^-Z; Tables 4,5,6, 7 for W^+W^-; Tables 8,9,10, 11 for ZZ.

IDh	Name	IDh	Name	IDh	Name	IDh	Name	IDh	Name	IDh	Name
44	UU	22	LL	25	LT	41	UM	14	MU	45	UT
52	TL	55	TT	24	LU	42	UL	24	LU	54	TU
42	UL	54	TU	45	UT	43	UP	34	PU	$\lambda_1\lambda_2$	

Table 1: Polarization ID (IDh) numbers and names for the V_1V_2 processes. Longitudinal polarization is denoted by the letter L and number 2, transverse polarization T and 5, unpolarized case U and 4, transverse left polarization M and 1, transverse right polarization P and 3. The first character is for the first gauge boson V_1, the second character for V_2. The last IDh denoted as $\lambda_1\lambda_2$ with $\lambda_i = 1, 2, 3$ is for other possibilities such as 12, 13, etc.

8.1 WZ

We first take the W^+Z process (IDp = 1) in Table 2. In order to obtain results at NLO QCD+EW one has to sum over all contributions with IDc from 1 to 17. Note that the special case of IDc = 0 is reserved for the unpolarized off-shell LO contribution. This is not needed for calculating polarized cross sections. It is however useful for crosschecking with other programs.

Using the provided python script `run_WZ.py` in `run_templates`, one calculates all those 17 contributions for all polarizations as follows:

```
CALCULATION = "POLAR_NLOQCDEW"
IDp_range = [1]
IDh_range = [22,55,52,25,44]
```

This script will create 17 output directories, and submit 17 jobs. After these jobs are finished, one then performs the sum over the results (either the integrated cross section or a given distribution) to obtain the desired NLO QCD+EW result. For the W^-Z process (see Table 3), one sets `IDp_range = [2]`. To submit jobs for both W^+Z and W^-Z processes, then `IDp_range = [1,2]`.

To obtain the off-shell LO results, one sets `CALCULATION = "off_shell"`. Other options including `"POLAR_LO"`, `"POLAR_NLOQCD"` are available as given in `run_WZ.py`.

8.2 W^+W^-

The processes and contributions for the W^+W^- case are provided in Tables 4, 5, 6, 7. They can all be calculated by invoking `python3 run_WpWm.py` with the following settings:

```
CALCULATION = "POLAR_NLOQCDEW"
# sub-dominant contributions for POLAR calculations:
gg_process = True
aa_process = True
bb_process = 2 # (choose 0 for NO bb, 1 for LO, 2 for NLO)
IDh_range = [22,55,52,25,44]
```

After all jobs are finished, one then combines the results as follows. For contributions in Tables 4, 5, 6, we add them together as usual (the errors are combined in quadrature). For the bottom-induced contributions in Table 7 we add the contributions with `IDc` from 1 to 17. Combining them all together gives the YesTW results in Ref. [17]. To obtain the NoTW results in Ref. [17] one then subtracts from the YesTW results the contributions with `IDc` from 18 to 21 in Table 7. Note that the sum from 1 to 17 in Table 7 includes the WW, tW and their interferences, while the contributions from 18 to 21 are the on-shell tW.

8.3 ZZ

The processes and contributions for the ZZ case are provided in Tables 8, 9, 10, 11. They can all be calculated by invoking `python3 run_ZZ.py` with the following settings:

```
CALCULATION = "POLAR_NLOQCDEW"
# sub-dominant contributions for POLAR calculations:
gg_process = True
bb_process = 2 # (choose 0 for NO bb, 1 for LO, 2 for NLO)
IDh_range = [22,55,52,25,44]
```

The $\gamma\gamma$ process contributes only to the off-shell case and is included there. The NLO QCD+EW cross sections are then obtained by adding all contributions.

8.4 Practical advice

The statistics for each contribution should be adjusted according to the importance of the contribution. The importance level is judged based on the value of the integrated cross section and the impact on the final distributions. For the VV at NLO QCD+EW, an optimal set of statistics for the up or down contributions from 1 to 17 reads

$$\log_2(N_{\mathrm{u,VV}}^{1\to17}) = \{24, 19, 19, 19, 24, 22, 22, 17, 17, 17, 23, 21, 21, 17, 21, 17, 21\}, \qquad (1)$$

where the LO term receives the highest statistics due to its large cross section. The {5,6,7} (for QCD) and {11,12,13,15,17} (for EW) contributions also receive significant statistics because of their impact on the distributions. There are significant fluctuations in the distributions of these contributions due to the cancellation nature of the dipole-subtraction method [20, 21]. To smooth out these fluctuations, additionally, one needs to combine results of different random seeds, more below.

For the gg and $\gamma\gamma$, a good choice is $N_{\mathrm{gg,VV}} = 2^{17}$ for both WW and ZZ.

For the b-induced contributions of the ZZ case one can choose $N_{\mathrm{b,ZZ}}^{1\to17} = N_{\mathrm{u,VV}}^{1\to17}/k$ with $k = 2$.

For the b-induced contributions of the WW case one can choose

$$\log_2(N_{\mathrm{b,WW}}^{1\to21}) = \{23, 18, 18, 18, 23, 23, 23, 16, 16, 16, 22, 22, 22, 16, 20, 16, 20, 23, 23, 21, 21\}, \qquad (2)$$

where the QCD contributions {6,7} and {18,19} receive large statistics because they are both huge (containing a top resonance) and cancel each other in the NoTW calculation. Similarly for the EW contributions {12,13} and {20,21}, with smaller statistics compared to their QCD counterparts due to smaller cross sections.

To further reduce fluctuations in the kinematic distributions, a good strategy is combining results of different random seeds (one can start with 5 seeds, and increase it if needed). Two common methods for combination are the normal average (using equal weights) and the weighted mean (using weights proportional to $1/\Delta_i^2$ with Δ_i being the error of the ith seed). The weighted-mean method gives smaller error, but it can only be trusted when there are no biases in the calculation of Δ_i. A good practice is to implement both methods and compare the results.

The standard way to calculate the scale uncertainties is to vary (μ_F, μ_R) around the central values (μ_{F0}, μ_{R0}), up and down by a factor of 2. A well-known method is the seven-point method. The seven scale points are $(\mu_F, \mu_R) = (i\mu_{F0}, j\mu_{R0})$ with $i = 0.5, 1, 2$ and $j = 0.5, 1, 2$, with the constraint of $i/j < 4$ and $j/i < 4$ to avoid wide-separated scales. In this calculation one can reduce significantly computing time by noting that only the QCD-correction contributions (`IDc=2,3,4,5,6,7` for all VV processes, for the case of WW there are additionally `IDc=18,19`) depend on both μ_F and μ_R. The other contributions (LO and NLOEW corrections) depend only on μ_F.

9 Guidelines for implementing new cuts

To implement a new kinematic cut, one first adds an entry to `cards/cuts.dat.in` to give it a name. Then edit the subroutine `Read_Cuts()` in `utilities/global_cuts.F90` to read in this value. Then edit the subroutine `DoCut_DiBoson()` in `utilities/Cut_tools.F` to implement this cut. For the NLOEW contributions with a photon radiation, one must do charged lepton photon recombination before applying the cut. Finally, one must run `make clean` and `make` in `tools/utilities` to update the `utilities` library.

A practical advice: if you are not sure about the momenta, you can use `vext_jend(5,nlegs_max)` to verify. `vext_jend` contains the information of the 4-momentum (first 4 entries) and particle-ID (fifth entry) of all external particles. The particle-IDs are given here [22], which is the standard particle numbering scheme for MC generators. Note that lepton-photon recombination has not been done on `vext_jend`.

10 Guidelines for implementing new distributions

To implement a new kinematic distribution, one first adds an entry to `utilities/Hist_init_VV.F` (with `VV=WW,WZ,ZZ`) to initialize a histogram template. Notice there that each histogram has an ID whose value is given by the variable `Ncout`. This ID will be used to fill the histogram in the corresponding `utilities/Hist_fill_VV.F`. As in the case of cuts, one must do charged lepton photon recombination before calculating the bin-position (`pos_bin`) for filling the histogram
(command `CALL XHFILL(ID,pos_bin,weight)`).

A List of processes and contributions

A.1 WZ

IDp	IDc	Name	Correction Type	Polarized
1	0	UDbarWpZ4lfull-born	LO	No
1	1	UDbarWpZ-born	LO	Yes
1	2	UDbarWpZ-virtQCD	NLOQCD	Yes
1	3	UDbarWpZrad-collQCD	NLOQCD	Yes
1	4	UDbarWpZind-collQCD	NLOQCD	Yes
1	5	UDbarWpZglu-fin	NLOQCD	Yes
1	6	DbargluUbarWpZ-fin	NLOQCD	Yes
1	7	UgluDWpZ-fin	NLOQCD	Yes
1	8	UDbarWpZ-virtEW	NLOEW	Yes
1	9	UDbarWpZrad-collEW	NLOEW	Yes
1	10	UDbarWpZind-collEW	NLOEW	Yes
1	11	UDbarWpZgam-fin	NLOEW	Yes
1	12	DbargamUbarWpZ-fin	NLOEW	Yes
1	13	UgamDWpZ-fin	NLOEW	Yes
1	14	UDbarWpZ-virtEW-decayZ	NLOEW	Yes
1	15	UDbarWpZgam-fin-decayZ	NLOEW	Yes
1	16	UDbarWpZ-virtEW-decayW	NLOEW	Yes
1	17	UDbarWpZgam-fin-decayW	NLOEW	Yes

Table 2: Process ID (IDp) and Contribution ID (IDc) numbers for the W^+Z process $pp \to e^+\nu_e\mu^+\mu^-$.

IDp	IDc	Name	Correction Type	Polarized
2	0	UbarDWmZ4lfull-born	LO	No
2	1	UbarDWmZ-born	LO	Yes
2	2	UbarDWmZ-virtQCD	NLOQCD	Yes
2	3	UbarDWmZrad-collQCD	NLOQCD	Yes
2	4	UbarDWmZind-collQCD	NLOQCD	Yes
2	5	UbarDWmZglu-fin	NLOQCD	Yes
2	6	UbargluDbarWmZ-fin	NLOQCD	Yes
2	7	DgluUWmZ-fin	NLOQCD	Yes
2	8	UbarDWmZ-virtEW	NLOEW	Yes
2	9	UbarDWmZrad-collEW	NLOEW	Yes
2	10	UbarDWmZind-collEW	NLOEW	Yes
2	11	UbarDWmZgam-fin	NLOEW	Yes
2	12	UbargamDbarWmZ-fin	NLOEW	Yes
2	13	DgamUWmZ-fin	NLOEW	Yes
2	14	UbarDWmZ-virtEW-decayZ	NLOEW	Yes
2	15	UbarDWmZgam-fin-decayZ	NLOEW	Yes
2	16	UbarDWmZ-virtEW-decayW	NLOEW	Yes
2	17	UbarDWmZgam-fin-decayW	NLOEW	Yes

Table 3: ID numbers for the W^-Z process $pp \to e^-\bar{\nu}_e\mu^+\mu^-$.

A.2 WW

IDp	IDc	Name	Correction Type	Polarized
3	0	UbarUWW4lfull-born	LO	No
3	1	UbarUWW-born	LO	Yes
3	2	UbarUWW-virtQCD	NLOQCD	Yes
3	3	UbarUWWrad-collQCD	NLOQCD	Yes
3	4	UbarUWWind-collQCD	NLOQCD	Yes
3	5	UbarUWWglu-fin	NLOQCD	Yes
3	6	UbargluUbarWW-fin	NLOQCD	Yes
3	7	UgluUWW-fin	NLOQCD	Yes
3	8	UbarUWW-virtEW	NLOEW	Yes
3	9	UbarUWWrad-collEW	NLOEW	Yes
3	10	UbarUWWind-collEW	NLOEW	Yes
3	11	UbarUWWgam-fin	NLOEW	Yes
3	12	UbargamUbarWW-fin	NLOEW	Yes
3	13	UgamUWW-fin	NLOEW	Yes
3	14	UbarUWW-virtEW-decayWm	NLOEW	Yes
3	15	UbarUWWgam-fin-decayWm	NLOEW	Yes
3	16	UbarUWW-virtEW-decayWp	NLOEW	Yes
3	17	UbarUWWgam-fin-decayWp	NLOEW	Yes

Table 4: ID numbers for the up-type quark induced W^+W^- processes $pp(U) \to e^+\nu_e\mu^-\bar{\nu}_\mu$. Each contribution includes u and c induced processes.

IDp	IDc	Name	Correction Type	Polarized
4	0	DbarDWW4lfull-born	LO	No
4	1	DbarDWW-born	LO	Yes
4	2	DbarDWW-virtQCD	NLOQCD	Yes
4	3	DbarDWWrad-collQCD	NLOQCD	Yes
4	4	DbarDWWind-collQCD	NLOQCD	Yes
4	5	DbarDWWglu-fin	NLOQCD	Yes
4	6	DbargluDbarWW-fin	NLOQCD	Yes
4	7	DgluDWW-fin	NLOQCD	Yes
4	8	DbarDWW-virtEW	NLOEW	Yes
4	9	DbarDWWrad-collEW	NLOEW	Yes
4	10	DbarDWWind-collEW	NLOEW	Yes
4	11	DbarDWWgam-fin	NLOEW	Yes
4	12	DbargamDbarWW-fin	NLOEW	Yes
4	13	DgamDWW-fin	NLOEW	Yes
4	14	DbarDWW-virtEW-decayWm	NLOEW	Yes
4	15	DbarDWWgam-fin-decayWm	NLOEW	Yes
4	16	DbarDWW-virtEW-decayWp	NLOEW	Yes
4	17	DbarDWWgam-fin-decayWp	NLOEW	Yes

Table 5: ID numbers for the down-type quark induced W^+W^- processes $pp(D) \to e^+\nu_e\mu^-\bar{\nu}_\mu$. Each contribution includes d and s induced processes.

IDp	IDc	Name	Correction Type	Polarized
5	1	glugluWW-loop	NNLO	Yes
7	1	gamgamWW-born	LO	Yes

Table 6: ID numbers for the gluon-gluon loop induced and photon-photon induced W^+W^- processes contributing to $pp \to e^+\nu_e\mu^-\bar{\nu}_\mu$.

A.3 ZZ

IDp	IDc	Name	Correction Type	Polarized
6	0	BbarBWW4lfull-born	LO	No
6	1	BbarBWW-born	LO	Yes
6	2	BbarBWW-virtQCD	NLOQCD	Yes
6	3	BbarBWWrad-collQCD	NLOQCD	Yes
6	4	BbarBWWind-collQCD	NLOQCD	Yes
6	5	BbarBWWglu-fin	NLOQCD	Yes
6	6	BbargluBbarWW-fin	NLOQCD	Yes
6	7	BgluBWW-fin	NLOQCD	Yes
6	8	BbarBWW-virtEW	NLOEW	Yes
6	9	BbarBWWrad-collEW	NLOEW	Yes
6	10	BbarBWWind-collEW	NLOEW	Yes
6	11	BbarBWWgam-fin	NLOEW	Yes
6	12	BbargamBbarWW-fin	NLOEW	Yes
6	13	BgamBWW-fin	NLOEW	Yes
6	14	BbarBWW-virtEW-decayWm	NLOEW	Yes
6	15	BbarBWWgam-fin-decayWm	NLOEW	Yes
6	16	BbarBWW-virtEW-decayWp	NLOEW	Yes
6	17	BbarBWWgam-fin-decayWp	NLOEW	Yes
6	18	BbargluBbarWW-fin-TW	NLOQCD	Yes
6	19	BgluBWW-fin-TW	NLOQCD	Yes
6	20	BbargamBbarWW-fin-TW	NLOEW	Yes
6	21	BgamBWW-fin-TW	NLOEW	Yes

Table 7: ID numbers for the bottom-quark induced W^+W^- processes $pp(B) \to e^+\nu_e\mu^-\bar{\nu}_\mu$.

B Details of contributions

B.1 `XX...4lfull-born`

This is the full off-shell calculation for the $XX \to 4$ leptons process at LO. PDFs are included. Calculation of unpolarized cross sections can be performed. Polarized cross sections are unavailable.

B.2 `XX...V1V2-born`

This is the DPA calculation for the $XX \to V_1V_2 \to 4$ leptons process at LO. PDFs are included. Calculation of polarized cross sections can be performed.

B.3 `XX...V1V2-virtQCD`

This is the DPA calculation for the QCD virtual corrections for the $XX \to V_1V_2$ process at NLOQCD. Fully leptonic decays are implicitly understood. PDFs are included. Calculation of polarized cross sections can be performed. The QCD virtual corrections here include virtual loop corrections and soft-gluon real corrections (called endpoint contribution in Ref. [21]). This contribution can be calculated using mass regularization or dimensional regularization. The default option is mass regularization.

IDp	IDc	Name	Correction Type	Polarized
8	0	UbarUZZ4lfull-born	LO	No
8	1	UbarUZZ-born	LO	Yes
8	2	UbarUZZ-virtQCD	NLOQCD	Yes
8	3	UbarUZZrad-collQCD	NLOQCD	Yes
8	4	UbarUZZind-collQCD	NLOQCD	Yes
8	5	UbarUZZglu-fin	NLOQCD	Yes
8	6	UbargluUbarZZ-fin	NLOQCD	Yes
8	7	UgluUZZ-fin	NLOQCD	Yes
8	8	UbarUZZ-virtEW	NLOEW	Yes
8	9	UbarUZZrad-collEW	NLOEW	Yes
8	10	UbarUZZind-collEW	NLOEW	Yes
8	11	UbarUZZgam-fin	NLOEW	Yes
8	12	UbargamUbarZZ-fin	NLOEW	Yes
8	13	UgamUZZ-fin	NLOEW	Yes
8	14	UbarUZZ-virtEW-decayZ2	NLOEW	Yes
8	15	UbarUZZgam-fin-decayZ2	NLOEW	Yes
8	16	UbarUZZ-virtEW-decayZ1	NLOEW	Yes
8	17	UbarUZZgam-fin-decayZ1	NLOEW	Yes

Table 8: ID numbers for the up-type quark induced ZZ processes $pp(U) \to e^+e^-\mu^+\mu^-$. Each contribution includes u and c induced processes.

IDp	IDc	Name	Correction Type	Polarized
9	0	DbarDZZ4lfull-born	LO	No
9	1	DbarDZZ-born	LO	Yes
9	2	DbarDZZ-virtQCD	NLOQCD	Yes
9	3	DbarDZZrad-collQCD	NLOQCD	Yes
9	4	DbarDZZind-collQCD	NLOQCD	Yes
9	5	DbarDZZglu-fin	NLOQCD	Yes
9	6	DbargluDbarZZ-fin	NLOQCD	Yes
9	7	DgluDZZ-fin	NLOQCD	Yes
9	8	DbarDZZ-virtEW	NLOEW	Yes
9	9	DbarDZZrad-collEW	NLOEW	Yes
9	10	DbarDZZind-collEW	NLOEW	Yes
9	11	DbarDZZgam-fin	NLOEW	Yes
9	12	DbargamDbarZZ-fin	NLOEW	Yes
9	13	DgamDZZ-fin	NLOEW	Yes
9	14	DbarDZZ-virtEW-decayZ2	NLOEW	Yes
9	15	DbarDZZgam-fin-decayZ2	NLOEW	Yes
9	16	DbarDZZ-virtEW-decayZ1	NLOEW	Yes
9	17	DbarDZZgam-fin-decayZ1	NLOEW	Yes

Table 9: ID numbers for the down-type quark induced ZZ processes $pp(D) \to e^+e^-\mu^+\mu^-$. Each contribution includes d and s induced processes.

IDp	IDc	Name	Correction Type	Polarized
10	1	`glugluZZ-loop`	NNLO	Yes
12	0	`gamgamZZ4lfull-born`	LO	No

Table 10: ID numbers for the gluon-gluon loop induced and photon-photon induced ZZ processes contributing to $pp \to e^+e^-\mu^+\mu^-$.

IDp	IDc	Name	Correction Type	Polarized
11	0	`BbarBZZ4lfull-born`	LO	No
11	1	`BbarBZZ-born`	LO	Yes
11	2	`BbarBZZ-virtQCD`	NLOQCD	Yes
11	3	`BbarBZZrad-collQCD`	NLOQCD	Yes
11	4	`BbarBZZind-collQCD`	NLOQCD	Yes
11	5	`BbarBZZglu-fin`	NLOQCD	Yes
11	6	`BbargluBbarZZ-fin`	NLOQCD	Yes
11	7	`BgluBZZ-fin`	NLOQCD	Yes
11	8	`BbarBZZ-virtEW`	NLOEW	Yes
11	9	`BbarBZZrad-collEW`	NLOEW	Yes
11	10	`BbarBZZind-collEW`	NLOEW	Yes
11	11	`BbarBZZgam-fin`	NLOEW	Yes
11	12	`BbargamBbarZZ-fin`	NLOEW	Yes
11	13	`BgamBZZ-fin`	NLOEW	Yes
11	14	`BbarBZZ-virtEW-decayZ2`	NLOEW	Yes
11	15	`BbarBZZgam-fin-decayZ2`	NLOEW	Yes
11	16	`BbarBZZ-virtEW-decayZ1`	NLOEW	Yes
11	17	`BbarBZZgam-fin-decayZ1`	NLOEW	Yes

Table 11: ID numbers for the bottom-quark induced ZZ processes $pp(B) \to e^+e^-\mu^+\mu^-$.

B.4 `XX...V1V2rad-collQCD`

This is the DPA calculation for the so-called collinear corrections of the real gluon emission process $XX \to V_1V_2g$ occurring at NLOQCD. PDFs are included. Calculation of polarized cross sections can be performed. This contribution can be calculated using mass regularization or dimensional regularization. The default option is mass regularization. This contribution is called the PK-operator contribution in [20]. The implementation of mass regularization is based on [21]. PDF counter terms are here included.

B.5 `XX...V1V2ind-collQCD`

This contribution is similar to the one described in Section B.4, but for the (anti)quark-gluon induced process $qg \to V_1V_2q'$.

B.6 `XX...V1V2glu-fin`

This is the DPA calculation for the remaining finite corrections of the real gluon emission process $XX \to V_1V_2g$ occurring at NLOQCD. PDFs are included. Calculation of polarized cross sections can be performed.

Dipole subtraction terms [21] are added here such that the integration can be performed in 4 dimensions using massless quarks.

B.7 `XbargluYbar...V1V2-fin`

This contribution is similar to the one described in Section B.6, but for the antiquark-gluon induced process $\bar{q}g \to V_1V_2\bar{q}'$.

B.8 `XgluY...V1V2-fin`

This contribution is similar to the one described in Section B.6, but for the quark-gluon induced process $qg \to V_1V_2q'$.

B.9 `XX...V1V2-virtEW`

This contribution is similar to the one described in Section B.3, but for the EW corrections to the production part (i.e. $XX \to V_1V_2$).

B.10 `XX...V1V2rad-collEW`

This contribution is similar to the one described in Section B.4, but for the EW corrections. The corrections here come from the real photon emission process $XX \to V_1V_2\gamma$. PDF counter terms are here included, using the DIS scheme or $\overline{\text{MS}}$ scheme provided in [23].

B.11 `XX...V1V2ind-collEW`

This contribution is similar to the one described in Section B.10, but for the (anti)quark-photon induced process $q\gamma \to V_1V_2q'$.

B.12 `XX...V1V2gam-fin`

This contribution is similar to the one described in Section B.6, but for the real photon emission process $XX \to V_1V_2\gamma$.

B.13 `XbargamYbar...V1V2-fin`

This contribution is similar to the one described in Section B.12, but for the antiquark-photon induced process $\bar{q}\gamma \to V_1V_2\bar{q}'$.

B.14 `XgamY...V1V2-fin`

This contribution is similar to the one described in Section B.13, but for the quark-photon induced process $q\gamma \to V_1V_2q'$.

B.15 `XX...V1V2-virtEW-decayVi`

This contribution is similar to the one described in Section B.9, but for the virtual EW corrections to the decay $V_i \to 2\,\text{leptons}$. The process is $XX \to V_1V_2 \to 4$ leptons.

B.16 `XX...V1V2gam-fin-decayVi`

This contribution is similar to the one described in Section B.12, but for the real photon emission off the decay $V_i \to 2\,\text{leptons}$. The process is $XX \to V_1V_2 \to 4\,\text{leptons} + \gamma$.

B.17 `BbargluBbarWW-fin-TW`

This is an on-shell tW contribution to the W^+W^- process, originated from the $\bar{b}g$ induced initial state. This contribution is needed to subtract the on-shell tW contribution occurring in the $\bar{b}g \to W^+W^-\bar{b}$ process (the `BbargluBbarWW-fin` contribution described in Section B.7). PDFs are included. Calculation of polarized cross sections can be performed. The process is $\bar{b}g \to W^+\bar{t} \to W^+W^-\bar{b}$. Details of this contribution are provided in [17].

B.18 `BgluBWW-fin-TW`

This contribution is similar to the one described in Section B.17, but originated from the bg induced process $bg \to W^-t \to W^+W^-b$. Details are also provided in [17].

B.19 `BbargamBbarWW-fin-TW`

This contribution is similar to the one described in Section B.17, but originated from the $\bar{b}\gamma$ induced process $\bar{b}\gamma \to W^+\bar{t} \to W^+W^-b$. Details are provided in [17].

B.20 `BgamBWW-fin-TW`

This contribution is similar to the one described in Section B.19, but originated from the $b\gamma$ induced process $b\gamma \to W^-t \to W^+W^-b$. Details are provided in [17].

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